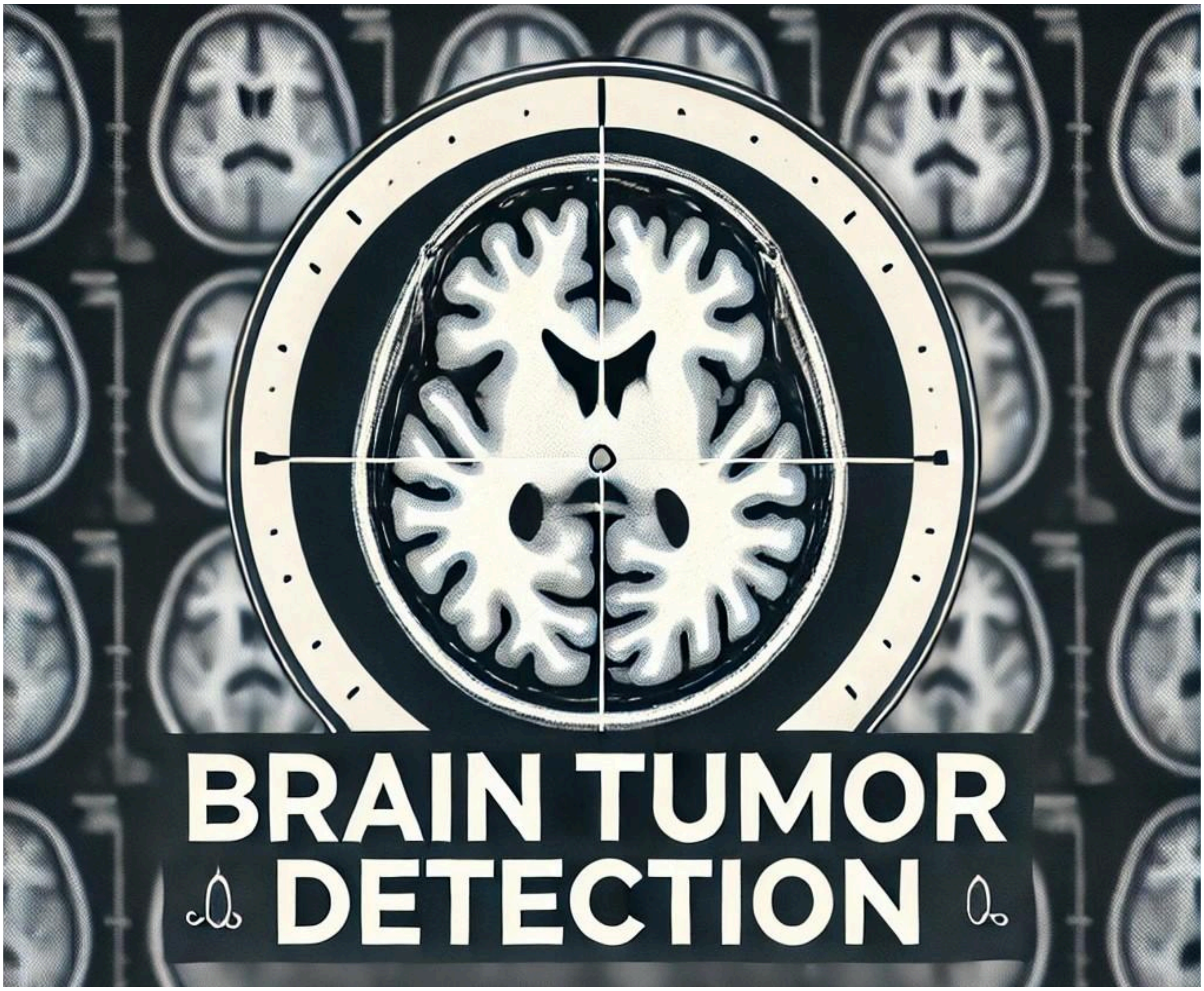
Brain Tumor Detection Report



Prepared by group 4:

- Nouar Chaima G1
- Hadidallah Meriem G2
- Ait mohammed Imene G2
- Farah Roumaissa G2

Introduction

Detecting brain tumors early is key to giving patients the best chance for successful treatment. MRI scans are commonly used to spot these tumors, but with so many images to analyze, it's easy for doctors to miss something. Manual review is also time-consuming and can lead to mistakes.

Machine learning and deep learning offer a solution by automating this process. These technologies can quickly and accurately analyze MRI scans, helping doctors detect brain tumors early. This project explores how we can use statistics, machine learning and deep learning to improve the diagnosis of brain tumors and support doctors in providing better care for their patients.

1. Dataset Preparation

○ Data Collection

The dataset used in this project consists of MRI images categorized into two classes: Tumor and No Tumor. The dataset initially contained a larger number of images; however, to balance the data and improve model performance, we reduced the dataset to 5000 images for each class (Tumor and No Tumor).

The dataset includes images in various formats and sizes, necessitating significant preprocessing to ensure consistency across the data. The images are primarily MRI scans, which are commonly used in brain tumor detection.

○ Data Preprocessing

Data preprocessing is a crucial step in preparing the dataset for analysis. Below are the key preprocessing steps performed:

- **Dataset Cleaning:** Balanced the dataset by selecting 5000 images for each class (Tumor and No Tumor).
- **Standardizing Image Formats:** All images were converted to the .jpg format for consistency.
- **RGB to Grayscale Conversion:** Converted images to grayscale to focus on intensity features.
- **Image Resizing:** Resize all images to 224 x 224 pixels for uniformity.

2. Methodology & Results

In this section, we focus on the use of statistical models, particularly the mean intensity-based approach, which is a traditional method for classifying images into tumor or non-tumor categories.

Statistical Model: Mean Intensity Classification

For the classic model, we used the mean intensity of the images as the primary feature. This method divides the image into smaller regions and computes the mean intensity value for each region. The classification of the image is based on the overall mean intensity across all regions, with a threshold applied to optimize the classification. By adjusting the threshold, we can fine-tune the decision boundary to improve accuracy.

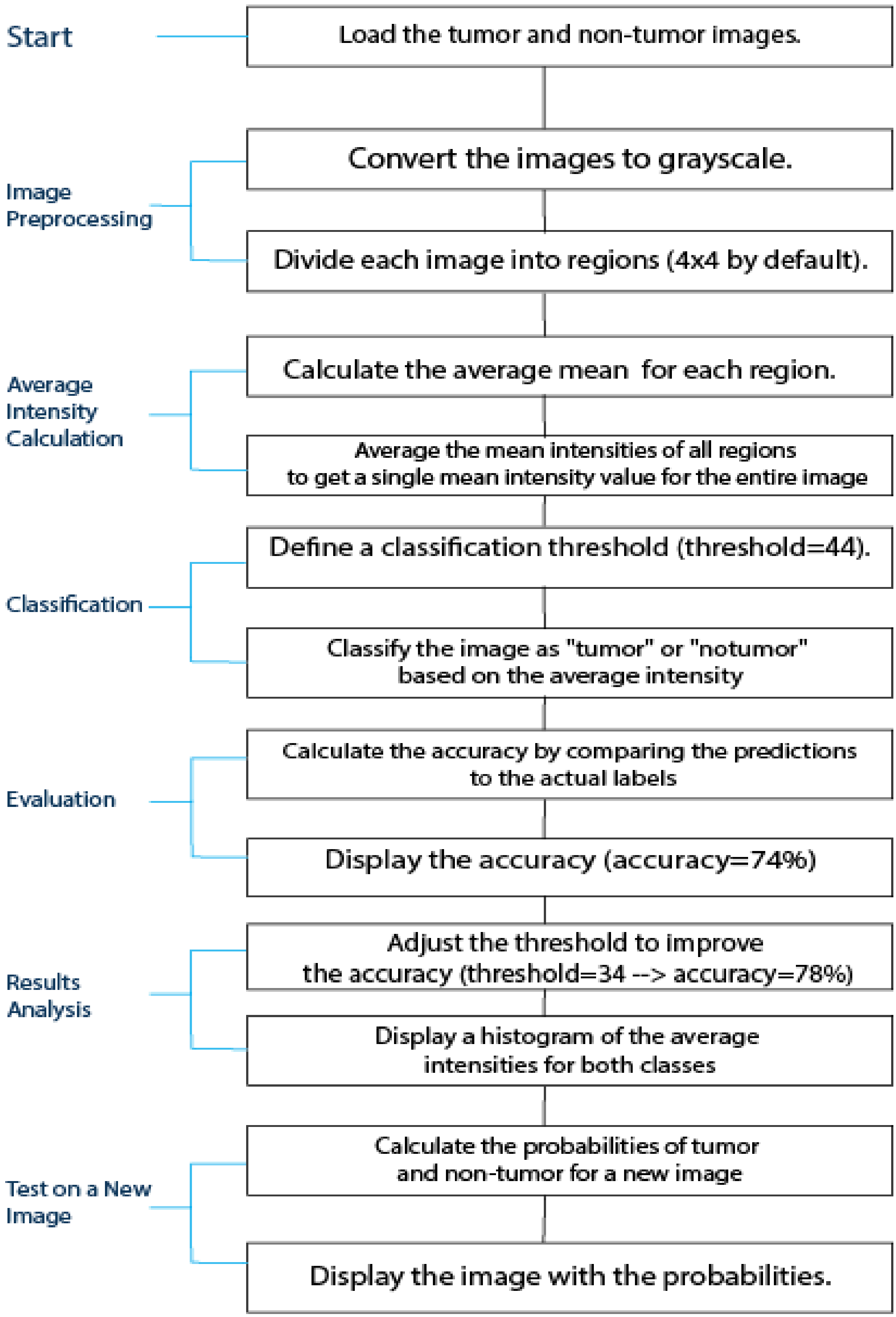
The steps are as follows:

1. **Image Preprocessing:**
   - Convert the image to grayscale.
   - Divide the image into smaller regions (default 4x4 grid).
2. **Mean Intensity Calculation:**
   - For each region, calculate the mean intensity of pixel values.
   - Average the mean intensities of all regions to get a single mean intensity value for the entire image.
3. **Classification Based on Threshold:**
   - A threshold is set based on the average mean intensity across the dataset. If the mean intensity of an image is below the threshold, the image is classified as a "tumor"; otherwise, it is classified as "notumor".

The threshold used for classification is set to the mean of the dataset's mean intensities, which allows for distinguishing tumor and non-tumor images.

figure 1: Flow Diagram for the Statistical Model

Flow diagram: Methodology and Results



3. Machine learning and Deep learning

After evaluating the performance of the statistical models, we explored more advanced approaches using Machine Learning and Deep Learning techniques.

3.1 Machine Learning Models

We used both **Support Vector Machine (SVM)** and **K-Nearest Neighbors (KNN)** models to classify brain tumor images. The features used for classification were extracted from the images, including **mean intensity**, **standard deviation**, and **entropy**, calculated over divided regions of the images.

Model Results:

The following table summarizes the results for both **SVM** and **KNN** models, including their performance metrics:

Model	Accuracy	Precision (Tumor)	Recall (Tumor)	F1-score (Tumor)	Precision (No Tumor)	Recall (No Tumor)	F1-score (No Tumor)	Confusion matrix
SVM	84.35%	0.79	0.93	0.85	0.92	0.76	0.83	[[920, 68], [245, 767]]
KNN	88.70%	0.86	0.91	0.89	0.91	0.86	0.89	[[904, 84], [142, 870]]

Analysis:

- **Accuracy:** KNN outperformed SVM with an accuracy of 88.70% vs 84.35%.
- **Precision and Recall:** SVM had strong recall for tumors (0.93) but lower recall for no tumors (0.76). KNN offered more balanced performance with higher precision and recall for both classes.
- **F1-Score:** KNN had better F1-scores (0.89 for both classes) compared to SVM (0.85 and 0.83), indicating better balance between precision and recall.

- **Confusion Matrix:** KNN had fewer false positives and false negatives than SVM, confirming its better overall performance.

3.2 Deep Learning Models

➢ Convolutional Neural Network (CNN)

Architecture and Layers:

1- Input Layer:

- **Shape**: 128x128 pixels, 3 color channels (RGB).

2- Convolutional Layers:

- **Layer 1**: 16 filters, 3x3 kernel, ReLU activation, followed by 2x2 MaxPooling.
- **Layer 2**: 32 filters, 3x3 kernel, ReLU activation, followed by 2x2 MaxPooling.

3- Fully Connected Layers:

- **Flatten**: Converts 2D feature maps to 1D vector.
- **Layer 3**: 128 units, ReLU activation.

4- Output Layer:

- 2 units, Softmax activation for binary classification (Tumor vs No Tumor).

Hyperparameters:

- **Epochs:** 20
- **Batch Size:** 32
- **Optimizer:** Adam optimizer with a learning rate of 0.001
- **Loss Function:** CrossEntropyLoss(for binary classification)
- **Metrics:** Accuracy

Training and Validation Results:

- **Test Accuracy:** 98.40%   (for 10 epochs: Test Accuracy: 97.95%)
- **Loss :** 0.0142

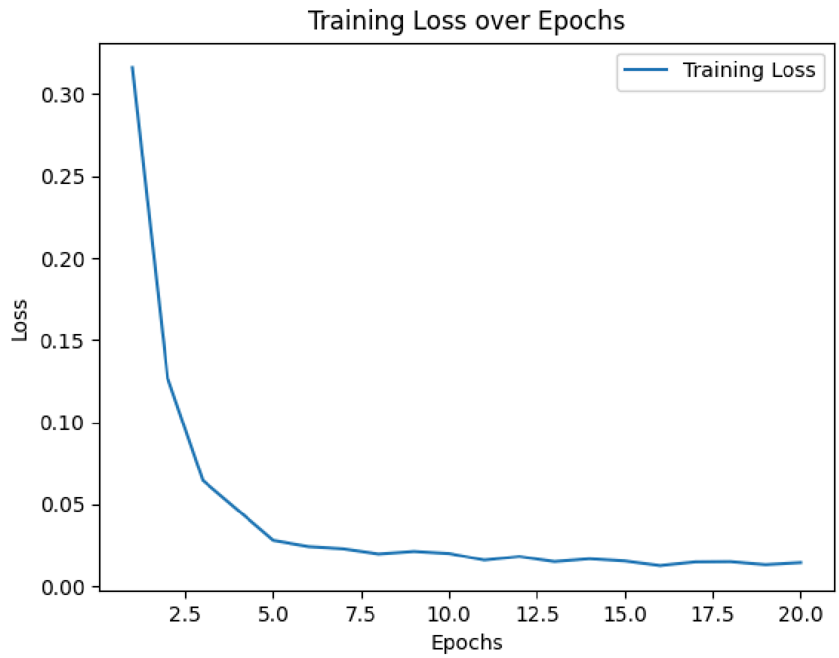


Figure 2: Loss Function Curve for CNN Model

➢ Artificial Neural Network (ANN)

Architecture and Layers:

Input Layer:

Input shape: 128x128 pixels, flattened to 16,384 features.

Hidden Layers:

- `fc1`: First hidden layer with 128 neurons, using ReLU activation and dropout for regularization.
- `fc2`: Second hidden layer with 64 neurons, using ReLU activation.

Output Layer:

- **`fc3`:** Final layer with 2 neurons for binary classification (Tumor vs No Tumor).

Hyperparameters:

- **Epochs:** 50
- **Batch Size:** 32

- **Optimizer:** Adam optimizer with a learning rate of 0.001
- **Loss Function:** CrossentropyLoss (for binary classification)
- **Metrics:** Accuracy

Training and Validation Results:

- **Test Accuracy:** 92.05% (for 30 epochs: Test Accuracy: 91.70%)
- **Loss:** 0.2866

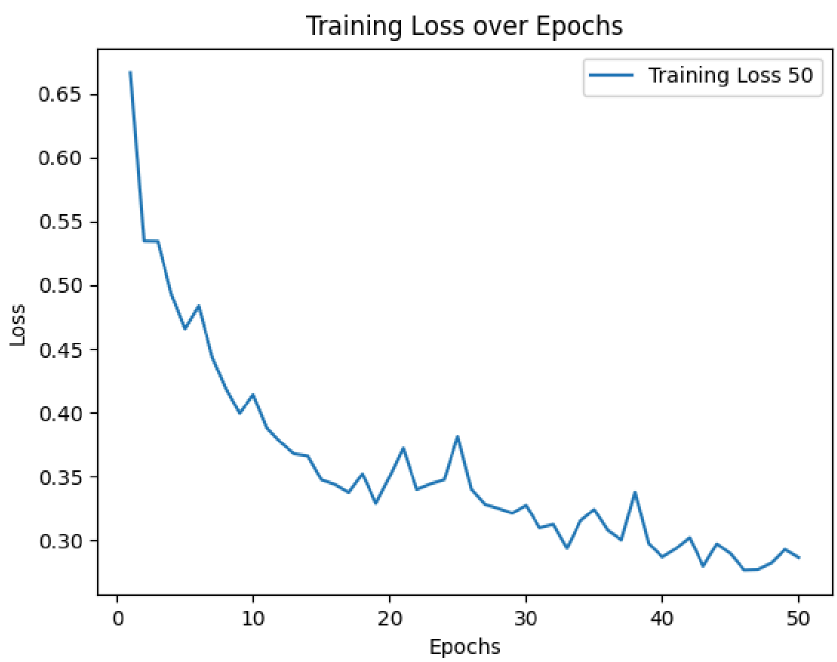


Figure 3: Loss Function Curve for ANN Model

Comparaison:

Aspect	**CNN**	**ANN**
Performance	Outperformed the ANN in terms of accuracy and loss. CNNs are better at handling image data.	Struggled with accuracy, lower performance due to inability to capture spatial patterns.

Feature Learning	Excelled at learning complex image features and spatial hierarchies.	Limited ability to capture spatial patterns in images, leading to poorer results.
Model Suitability	More suitable for image classification tasks as it processes spatial information effectively.	Less suitable for image classification tasks due to lack of spatial information processing.

Conclusion:

In this project, we evaluated several models for brain tumor image classification, including statistical models, machine learning (SVM and KNN), and deep learning models (CNN and ANN).

- The **statistical model** based on **mean intensity** provided a basic classification but lacked the accuracy and feature extraction capabilities of more advanced models.
- Among the **machine learning models**, the **KNN** classifier outperformed SVM in terms of both accuracy and precision, achieving better overall classification performance. **SVM** showed strong recall for tumors but struggled with the no-tumor class.
- The **deep learning models** showed the most promising results. The **CNN** outperformed both the machine learning models and the ANN, owing to its ability to automatically learn hierarchical features from image data. The **ANN**, while effective, had limitations in capturing spatial patterns compared to CNN, resulting in slightly lower performance.

Ultimately, the **deep learning-based CNN model** proved to be the most effective approach for this classification task, showing superior accuracy and the ability to handle complex spatial information in the images.